

Article

On the Different Fair Allocations of Economic Benefits for Energy Communities

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Abstract: Energy Communities (ECs) are aggregations of users that cooperate to achieve economic benefits by sharing energy instead of operating individually in the so-called “disagreement” case. As there is no unique notion of fairness for the cost/profit allocation of ECs, this paper aims to identify an allocation method that allows for an appropriate weighting of both the interests of an EC as a whole and those of all its members. The novelty is in comparing different optimization approaches and cooperative allocation criteria, satisfying different notions of fairness, to assess which one may be best suited for an EC. Thus, a cooperative model is used to optimize the operation of an EC that includes two consumers and two solar PV prosumers. The model is solved by the “Social Welfare” approach to maximizing the total “incremental” economic benefit (i.e., cost saving and/or profit increase) and by the “Nash Bargaining” approach to simultaneously maximize the total and individual incremental economic benefits, with respect to the “disagreement” case. Since the “Social Welfare” approach could lead to an unbalanced benefit distribution, the Shapley value and Nucleolus criteria are applied to re-distribute the total incremental economic benefit, leading to higher annual cost savings for consumers with lower electricity demand. Compared to “Social Welfare” without re-distribution, the Nash Bargaining distributes 39–49% and 9–17% higher annual cost savings to consumers with lower demand and to prosumers promoting the energy sharing within the EC, respectively. However, total annual cost savings drop by a maximum of 5.5%, which is the “Price of Fairness”.



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1. Introduction

Promoting a better match of the local energy demand with energy produced from Renewable Energy Sources (RESs) is fundamental to achieving a greener, more sustainable and fairer energy system in the future, thus mitigating the harmful effects of climate change and reducing the occurrence of energy crises caused by geopolitical events [1]. With this aim, the European Union introduced the regulatory tool of the Energy Community (EC) [2] by issuing the Renewable Energy Directive (RED II) [3] and the Internal Electricity Market Directive (IEMD) [4]. An EC is an aggregation of energy users from different consumption sectors (e.g., residential, commercial, tertiary, etc.) that can share renewable energy among themselves [5], thereby contributing to the achievement of the environmental targets set out in the “Fit for 55” plan of the European Commission (e.g., emission reduction of at least 55% within 2030 compared to the levels of 1990) [6]. ECs could lead to economic (e.g., cost savings) [7], environmental (e.g., emission reductions) [8] and social (e.g., access to distributed RES for vulnerable users) [9] benefits to its members.

1.1. Literature Review

The literature on ECs has mainly focused on the following two problems: (i) the definition of the type and number of users and the choice of the optimal sizes and operation of the energy conversion and storage technologies, and (ii) the fair allocation of the economic benefits (i.e., costs or profits) of an EC to its members. This work embraces both these research areas, considering the operation optimization of an EC composed of energy users from different consumption sectors, and the allocation of the optimal economic benefit of the EC to its members, with a particular insight into the concept of fairness associated with different allocation methods.

Works dealing with problem (i) mainly use linear programming (LP) or mixed-integer linear (non-linear) programming (MILP or MINLP) approaches to model and optimize different energy systems of the EC [10–16]. Chang et al. [10] minimized the cost of buying electricity from the grid for an EC and proposed a methodology based on K-means clustering to allocate different options of community energy storage among the households of the EC. Cutore et al. [11] optimized the design and operation of a residential EC in Italy to maximize the net-present value (including investment and operational costs) and, then, analyzed its energy, environmental and social performance by calculating the total self-consumption and self-sufficiency, the avoided carbon dioxide emissions and an energy poverty index, respectively. Lazzari et al. [12] developed a multi-objective optimization framework to find the optimal combination of users within an EC and the optimal allocation of solar energy generation to users. Barone et al. [13] developed a bi-level approach to optimally select the type and number of users of an EC based on a Peer-to-Peer (P2P) energy-sharing mechanism, where users share energy at cheaper prices compared to the grid ones. The self-sufficiency of subgroups of users and the self-consumption of the whole EC are maximized in the first and second levels, respectively. Belloni et al. [14] optimized the sizes of solar PV and electrical energy storage systems in an EC located in Italy, considering several scenarios characterized by different locations (e.g., northern, central and southern Italy), maximum size of PV plants and market prices associated with the pre- and post-COVID-19 pandemic. Dal Cin et al. [15] conducted a multi-objective optimization to find the optimal design and operation of different EC configurations by minimizing their total costs and CO₂ emissions. Stegen et al. [16] presented a two-step approach to first maximize the economic profits of an EC and then the minimum profit that each user can obtain by entering the EC compared to its independent operation. The results of the previous works have shown that the cooperation of energy users from the residential sector only [10–13] or from both the residential and non-residential sectors [14–16] can lead to economic benefits for ECs. The number of users (consumers and/or prosumers) within these ECs varies approximately between 2 and 40, with solar PV systems (i.e., the most used technology) ranging in size from 3–10 kW in households to 50–500 kW in buildings with other end-uses (e.g., schools and industries). However, it is worth highlighting that the works cited above did not consider any allocation of the total cost/profit of the EC to its members.

Works dealing with problem (ii) evaluate different strategies to allocate the total cost/profit of an EC to its members. A first strategy consists of establishing local energy markets based on different pricing mechanisms. Li and Okur [17] applied uniform pricing, time-of-use energy pricing and segmented energy pricing to distribute the optimal operational cost of an EC. Volpato et al. [18] presented an allocation mechanism that encourages users who have the higher economic benefit of moving from the status of independent consumer to that of independent prosumer to become members of an EC. Casalicchio et al. [19] allocated the total cost saving of a residential EC by applying a mechanism that evaluates the contribution of each member to the cost saving of the whole community. Zheng et al. [20] found that a P2P energy-sharing mechanism within an EC could minimize the total operational cost by 24.6% compared to a Peer-to-Grid (P2G) operation where each member exchanges energy only with the electric grid. Fina [21] allocated the grid costs of an EC by applying the “income”, “contribution” and “proximity” based re-distribution

mechanisms, which benefit low-income users, users with EC assets and users sharing energy locally, respectively. Although the cost/profit allocation obtained by establishing local energy markets is a valid practical option [17,22], it could not guarantee fairness and the property of “efficiency”, i.e., the equality between the optimal economic benefits of the system and the sum of the allocated individual benefits [23], which is required to guarantee energy democracy and justice in ECs [24]. In practice, in local energy markets, it is difficult to fulfill the “efficiency” property because energy prices are usually set before the optimal operation of the system is obtained and, consequently, the individual benefits (depending on the energy prices defined) are distributed before the optimal economic benefits of the system are known.

Another strategy to allocate the total cost/profit of an EC to its members is that of using cooperative game theory approaches [25], where users work together to achieve a common goal for the EC, which is in line with the European directives to foster a sustainable low-carbon energy transition [26]. Therefore, only cooperative modeling approaches are analyzed in this work, while non-cooperative approaches, which model users acting on their own and pursuing individual goals selfishly, are usually not suitable for ECs [27]. In cooperative models, the main issue is to define how the optimal economic benefit of the EC should be distributed among users by choosing a possibly fair method [19,20,28] that encourages users to participate in the EC. It is worth highlighting that there are different notions of fairness associated with cooperative approaches and that there is not a unique approach to ensure fairness in the cost/profit allocation of an EC. The Shapley value and Nucleolus are cooperative allocation criteria that fairly re-distribute the total economic benefits found by the “Social Welfare” (SW) optimization, which aims to minimize (maximize) the total costs (profits) of a system while usually neglecting the individual benefits of users [29]. The Shapley value criterion [30], according to the notion of “individual fairness” [31], allows for the allocation of higher profits or lower costs to members of the EC who contribute more to increasing the economic benefits of the coalitions of the EC (i.e., groups of users within the EC). The Nucleolus criterion [32], according to the notion of “collective fairness” [31], guarantees that each coalition of users has a higher economic benefit inside the EC compared to the case outside the EC. This prevents users from having an incentive to leave the EC, thus ensuring its “stability”, because each coalition of users will always find it economically convenient to participate in the EC. The Nash Bargaining (NB) optimization approach [33], according to the notion of “proportional fairness”, allows for the simultaneous optimization of the total benefit of the EC and the individual benefits of its members compared to a “disagreement” case in which each member acts individually (i.e., without cooperation).

The following works have applied the Shapley value and Nucleolus criteria and the NB optimization approach to allocate the total benefits of an EC. Vespermann et al. [34] used different cooperative allocation criteria for an EC and analyzed their properties in terms of efficiency, individual rationality (i.e., each user has a higher benefit from participating in the EC compared to its independent operation) and stability. Volpato et al. [35] applied the Shapley value to distribute the optimal daily economic benefit of an EC in Italy, resulting in higher rewards for prosumers than consumers given their higher contributions to the optimal economic benefit of the system. Zatti et al. [28] optimized the capacities of different generation and storage units within an Italian EC comprised of commercial and residential users and then applied the Shapley value to allocate the total economic benefit. Cremers et al. [36] reviewed different applications of the Shapley value for ECs, also proposing a new method (called “Stratified expected value”) to approximate the calculation of the Shapley value in the case of communities of up to 200 prosumers. Pedrero et al. [23] proposed the novel “Nested Shapley value” criterion, which assigns users to pre-defined clusters, to fairly distribute the investment and operational costs of a multi-dwelling building composed of 250 apartments. Siqin et al. [37] formulated a distributionally robust optimization model to optimize the operation of a multi-EC system. An improved Shapley value criterion was proposed to allocate higher profits to the participants with higher solar PV consumption. Yang et al. [38] applied the Nucleolus criterion to distribute the

daily investment and operational costs of an EC by minimizing the users' "dissatisfaction" for each coalition (i.e., group of users), which is calculated as the difference between the benefits of the independent coalition and the benefits of the same coalition within the EC. Bossu et al. [31] modified the Shapley value and Nucleolus criteria to specifically reward the flexibility and energy savings of users within an EC compared to their individual operation. Fioriti et al. [39] optimized the design and operation of an EC and then applied different cooperative allocation criteria such as Shapley value, Nucleolus and the hybrid "Shapley-Nucleolus", the latter ensuring a trade-off between the "individual fairness" of the Shapley value and the stability of the Nucleolus. Limmer [40] analyzed how the fairness (according to the notion of "individual fairness" of the Shapley value) and the stability of the cost allocation by Shapley value and Nucleolus evolve over time for a real EC. As the number of households in the EC increased, the Shapley value and Nucleolus became more unstable and fairer, respectively. Fischer and Toffolo [41] used an NB model to maximize the economic gains of different energy consumers and a producer that cooperate in a municipality compared to the case without cooperation. Ding et al. [42] solved an NB problem to fairly allocate P2P flows of electricity, heat, and hydrogen among different agents of a multi-energy system. Li et al. [43] developed an NB model to fairly allocate the total investment and operational costs of an EC to its users based on their contribution to storage sharing. Devi et al. [44] presented a multi-stage NB model that includes the definition of the optimal trading price in a smart grid composed of ten consumers and ten prosumers.

From the literature analyzed above [23,28,31,35–37,39,40], several papers used the Shapley value to attempt to fairly allocate the total economic benefits of an EC to its members. In addition, some papers assumed a specific notion of fairness (e.g., "individual fairness" in [39] and "collective fairness" in [38]), without considering that this choice affects the economic benefits allocated to the EC members. As highlighted in [36], the literature lacks more in-depth discussions on the fairness of the cost/profit distribution of ECs, which needs to be further investigated by applying not only one but several cooperative approaches, as there is not a unique definition of fairness.

1.2. Objective and Novelty

It clearly emerges that the literature lacks a comprehensive analysis of the distribution of the economic benefits among EC members by applying and comparing different optimization approaches (e.g., SW and NB) and cooperative allocation criteria (e.g., Shapley value and Nucleolus) that adhere to different notions of fairness [16]. This paper aims to fill this gap by identifying, among those proposed in the literature, an allocation method that allows for an appropriate weighting of both the interests of an EC as a whole and those of all its members, as well as between those of the energy consumers and prosumers. To this end, different possible solutions to the problem of fair allocation of economic benefits for an EC are obtained, taking into account different notions of fairness. The novelty is indeed the comprehensive comparison between the solutions obtained by applying different optimization approaches and cooperative allocation criteria that satisfy different notions of fairness, such as "individual", "collective" and "proportional" fairness, to assess which method may be best suited to achieve an optimal compromise of the interests of all EC members.

This paper applies different optimization approaches and cooperative allocation criteria to distribute the optimal economic benefits of an EC among its members. A cooperative model is used to optimize the operation of a real-world setting EC, which includes two consumers from the residential and commercial sectors and two prosumers equipped with solar Photovoltaic (PV) plants from the residential and agricultural sectors. A SW optimization is used to maximize only the total incremental economic benefits of the EC, while an NB optimization is used to maximize the total and individual incremental economic benefits simultaneously. The incremental benefits (i.e., cost savings and/or profit increases) are calculated on an annual basis compared to a reference "disagreement" case in which each user acts independently and individually. As the SW optimization could lead to an

inequitable distribution of individual economic benefits, the Shapley value and Nucleolus criteria are applied a posteriori to re-distribute the optimal economic benefit of the EC fairly. To the best of the authors' knowledge, this paper is the first to compare different solutions of fair cost/profit allocation for an EC obtained by applying the Shapley value and Nucleolus criteria (after the SW optimization), which satisfy the "individual fairness" and "collective fairness" respectively, and by carrying out the NB optimization, which satisfies "proportional fairness". In summary, this paper contributes to the current literature on the fair allocation of economic benefits for ECs by comparing three distributions based on different notions of fairness and by identifying among them the most appropriate one for an EC composed of both consumers and prosumers. The paper is organized as follows. Section 2 presents the methodology. Section 3 presents the results. Section 4 summarizes the main findings and conclusions.

2. Materials and Methods

Section 2.1 presents the considered EC and the cooperative model for its optimization. Section 2.2 presents the "Social Welfare" (SW) and the "Nash Bargaining" (NB) optimization approaches that are used to solve the cooperative model of the EC. Section 2.3 describes the cooperative criteria of Shapley value and Nucleolus to fairly re-distribute the optimal economic benefit of the EC found by the SW optimization. Section 2.4 shows the input data of the optimization model.

2.1. Cooperative Optimization Model of the EC

Figure 1 shows the EC under analysis, which consists of commercial (Com) and residential (Res1) consumers, and agricultural (Agr) and residential (Res2) prosumers equipped with a PV plant. This EC represents a general real-world setting as it contains the two main types of energy end-users, i.e., consumers and prosumers, with energy demand profiles representative of different consumption sectors. The electricity demand of each user is considered flexible, in particular, it can be optimally shifted in each hour through a Price-Based Demand Response (PBDR) program with a Real-Time Pricing (RTP) strategy [45], which sets daily profiles of both Peer-to-Grid (P2G) purchase and sale prices according to the daily profile of the day-ahead market price (see Section 2.4). In addition, the energy sharing within the community is promoted by a Peer-to-Peer (P2P) mechanism [42,46] with a simplified version of the mid-market pricing scheme [47], which ensures cheaper P2P prices compared to the P2G ones. Thus, in this energy and pricing framework, an EC member with energy deficit (e.g., consumer and prosumer during low-null PV production) can purchase energy from members with an energy surplus (i.e., realizing a P2P energy exchange) or from the external electric distribution grid (i.e., realizing a P2G energy exchange). In a similar way, an EC prosumer with an energy surplus can sell energy to members with energy deficit or to the external electric grid.

A cooperative optimization model of the EC is used to maximize the incremental economic benefits (i.e., cost savings for consumers and cost savings plus profit increases for prosumers) compared to a "disagreement" case (considered as reference), in which users do not cooperate and only exchange energy with the electric distribution grid independently (i.e., at P2G prices). The operation optimization of the EC is carried out with an annual time horizon, which is represented by 8 typical days of global solar irradiance, user electricity demands and electricity prices found by applying a K-means clustering algorithm [48]. According to [49,50], each typical day is selected as the real day with the lowest value of the Euclidean distance from the centroid of the associated cluster.

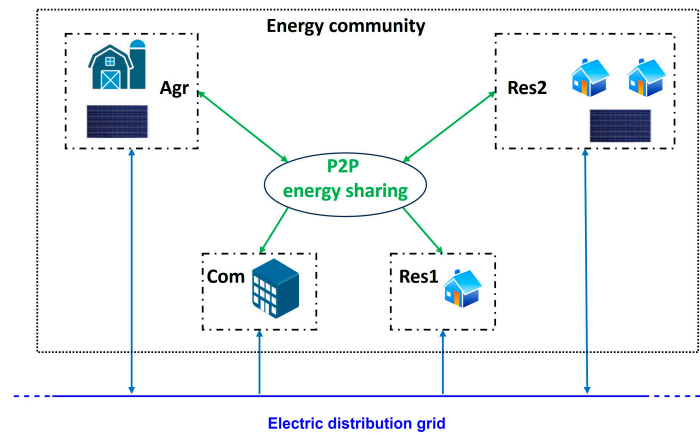


Figure 1. Energy community with commercial (Com) and residential (Res1) consumers, and agricultural (Agr) and residential (Res2) prosumers. The blue and green arrows represent the energy exchanged with the electric distribution grid (i.e., P2G energy) and the energy shared among users (i.e., P2P energy), respectively.

It should be noted that the cooperative model is formulated and solved by neglecting the uncertainty in solar irradiance, electricity demands and grid electricity prices, which usually depend on intermittent weather conditions, variable user habits and unpredictable socio-economic conditions, respectively. The reason for this choice is to assess only the impact of the optimization approaches (Section 2.2) and the cooperative allocation criteria (Section 2.3) on the allocation of incremental economic benefits.

The constraints of the model include the indices c , j , i , k , and t , which identify a consumer, a prosumer, a general member of the EC (consumer c or prosumer j), a typical day of the year and a specific hour of the day, respectively. The decision variables (defined for each hour of each typical day) are associated with the operation of the EC and include the P2G energy flows ($p_{i,k,t}^{imp}$ for import and $p_{j,k,t}^{exp}$ for export), the P2P energy flows ($p_{i,k,t}^{-}$ for purchase and $p_{j,k,t}^{+}$ for sale) and the flexible electricity demands ($D_{i,k,t}^{shift}$). The sizes of the PV plants of the two prosumers are instead fixed (Section 2.4). The variables associated with the energy flows and the economic benefits have kWh and € as units of measurement, respectively.

The total electricity balance of the EC is as follows:

$$\left(\sum_i p_{i,k,t}^{imp} - \sum_j p_{j,k,t}^{exp} \right) + \sum_j PV_{j,k,t} - \sum_i D_{i,k,t}^{shift} = 0 \quad (1)$$

where $p_{i,k,t}^{imp}$, $p_{j,k,t}^{exp}$, $PV_{j,k,t}$ and $D_{i,k,t}^{shift}$ are the P2G energy imported, the P2G energy exported (only for a prosumer j), the energy generated by the PV plant (only for a prosumer j) and the electricity demand of an EC member after a possible hourly shift of the input electricity demand due to the PBDR application, respectively.

The individual electricity balances of a consumer c (Equation (2)) and a prosumer j (Equation (3)) are as follows:

$$p_{c,k,t}^{imp} + p_{c,k,t}^{-} - D_{c,k,t}^{shift} = 0 \quad (2)$$

$$\left(p_{j,k,t}^{imp} - p_{j,k,t}^{exp} \right) + \left(p_{j,k,t}^{-} - p_{j,k,t}^{+} \right) + PV_{j,k,t} - D_{j,k,t}^{shift} = 0 \quad (3)$$

where $p_{c,k,t}^{-}/p_{j,k,t}^{-}$ and $p_{j,k,t}^{+}$ are the P2P energy purchased by consumer/prosumer c/j and the P2P energy sold by prosumer j , respectively. Notice that the P2P energy flows represent the energy shared within the EC and, therefore, appear only in the individual energy balances (Equations (2) and (3)), but not in the total energy balance of the EC (Equation (1)).

Moreover, according to Equations (1)–(3), the sum of the P2P energy flows (purchase and sale) across all members i in each time step t results to be zero.

The total energy exchanged between the EC and the electric distribution grid (imported and exported in Equations (4) and (5), respectively) is constrained by the grid capacity P^{max} :

$$\sum_i p_{i,k,t}^{imp} \leq P^{max} \quad (4)$$

$$\sum_j p_{j,k,t}^{exp} \leq P^{max} \quad (5)$$

In a similar way, assuming that the P2P energy exchanges rely on a local grid in which the EC operates, the total energy shared within the EC (which depends on the sum of $p_{i,k,t}^-$ over all users i and the sum of $p_{j,k,t}^+$ over all prosumers j) is limited by P^{max} as well.

The PBDR constraints to optimally shift the user electricity demands are as follows:

$$\sum_{t=1}^{24} D_{i,k,t}^{shift} = \sum_{t=1}^{24} D_{i,k,t} \quad (6)$$

$$D_{i,k}^{min} \leq D_{i,k,t}^{shift} \leq D_{i,k}^{max} \quad (7)$$

$$(1 - D^{var}) \cdot D_{i,k,t} \leq D_{i,k,t}^{shift} \leq (1 + D^{var}) \cdot D_{i,k,t} \quad (8)$$

where $D_{i,k,t}$, $D_{i,k}^{max} / D_{i,k}^{min}$ and D^{var} are the input electricity demand (i.e., before the operation optimization of the EC and therefore the PBDR application), the daily maximum/minimum of the input demand and the hourly maximum fraction of the load that can be shifted (the value assumed here is 0.1), respectively. Equation (6) states that the daily electricity demand after the PBDR is unchanged compared to the daily input demand. Constraint (7) states that the hourly electricity demand after the PBDR ($D_{i,k,t}^{shift}$) cannot be higher/lower than the daily maximum/minimum of the input demand. The upper bound ($D_{i,k}^{max}$) is required to avoid that the prosumer demand after PBDR exceeds the available PV energy generated, which is known since the PV surfaces are fixed as parameters and do not represent decision variables (Section 2.4). This guarantees the energy self-sufficiency of each prosumer in the middle hours of each typical day k , i.e., its energy demand is satisfied using the PV energy generated [51]. Moreover, the lower bound ($D_{i,k}^{min}$) is utilized to prevent users from experiencing energy discomfort when PBDR significantly reduces the load [52,53]. Constraint (8) defines the range of variation of $D_{i,k,t}^{shift}$ in each hour t of each day k according to the hourly maximum fraction of the load that can be shifted. It is worth highlighting that the total electricity demand of the EC users is considered sufficient to optimize the operation of the EC and to allocate the optimal total economic benefit to its members, which is the focus of this work. Accordingly, it is beyond the scope of this work to go into the details of the electricity consumption profiles of each EC member. A more detailed analysis of these electricity demands, even modified in accordance with the PBDR program, would require taking into account the specific electric appliances of each user and optimizing the associated on/off operational state by means of additional binary decision variables, thus increasing the complexity of the cooperative model of the EC.

The economic benefits of users within the EC are calculated for each day k :

$$u_{i,k}^- = \sum_{t=1}^{24} p_{i,k,t}^- \cdot c_{k,t}^{tr,b} + p_{i,k,t}^{imp} \cdot c_{k,t}^{imp} \quad (9)$$

$$u_{j,k}^+ = \sum_{t=1}^{24} p_{j,k,t}^+ \cdot c_{k,t}^{tr,s} + p_{j,k,t}^{exp} \cdot c_{k,t}^{exp} \quad (10)$$

where $u_{i,k}^-$, $u_{j,k}^+$, $c_{k,t}^{tr,b}/c_{k,t}^{tr,s}$ and $c_{k,t}^{imp}/c_{k,t}^{exp}$ are the daily cost, the daily profit (only for a prosumer j), the P2P and the P2G purchase/sale prices, respectively. It should be noted that the P2G purchase price (i.e., $c_{k,t}^{imp}$) is obtained from the P2G sale price (i.e., $c_{k,t}^{exp}$) by adding an “external” grid tariff, while the P2P prices are calculated in the interval defined by the P2G purchase and sale prices (see Section 2.4). Similar to the P2G purchase price, the P2P purchase price can also be split into the P2P sale price plus an “internal” grid tariff, which represents the costs of the local grid in which the EC operates. However, this “internal” grid tariff is lower than the “external” grid tariff (hereafter referred to as the grid tariff for simplicity) to promote the energy sharing within the EC and thus reward the avoided use of the external grid (the use of which would incur higher costs for the energy distribution than for the energy shared locally within the EC).

The incremental economic benefits of users derived from their cooperation within the EC compared to the disagreement case are calculated for each day k :

$$u_{i,k}^{incr-} = u_{i,k}^{dis-} - u_{i,k}^- \quad (11)$$

$$u_{i,k}^{incr-} \geq 0 \quad (12)$$

$$u_{j,k}^{incr+} = u_{j,k}^+ - u_{j,k}^{dis+} \quad (13)$$

$$u_{j,k}^{incr+} \geq 0 \quad (14)$$

where $u_{i,k}^{incr-}$ and $u_{j,k}^{incr+}$, $u_{i,k}^{dis-}$ and $u_{j,k}^{dis+}$ are the incremental cost benefit (i.e., cost saving) of a member i and the incremental profit benefit (i.e., profit increase) of a prosumer j obtained by moving from the disagreement case to the cooperative case within the EC, the daily cost of a member i and the daily profit of a prosumer j in the disagreement case, respectively. It should be noted that $u_{i,k}^{dis-}$ and $u_{j,k}^{dis+}$ are computed by solving an operation optimization for each user independently. It is assumed that even in the disagreement case, each user demand can be optimally shifted according to the same PBDR program used in the cooperative case. Constraints (12) and (14) are required to guarantee that users obtain higher (or at least equal) economic benefits by their cooperation within the EC (i.e., lower costs, Equation (9), and higher profits, Equation (10)) than by their individual operation, thus providing a reason for the formation of the EC and for the users to accept being part of the community.

Since consumers can only buy P2G and P2P energy, but prosumers can either buy or sell P2G and P2P energy (depending on whether they have an energy deficit or surplus, respectively, Equation (3)), the daily incremental economic benefits of a consumer c (Equation (15)) and a prosumer j (Equation (16)) become the following:

$$u_{c,k}^{incr} = u_{c,k}^{incr-} \quad (15)$$

$$u_{j,k}^{incr} = u_{j,k}^{incr-} + u_{j,k}^{incr+} \quad (16)$$

Eventually, the annual incremental economic benefit of a member i is as follows:

$$u_i^{incr} = \sum_k u_{i,k}^{incr} \cdot w_k \quad (17)$$

where $u_{i,k}^{incr}$ is equal to $u_{c,k}^{incr}$ for a consumer c (Equation (15)) and $u_{j,k}^{incr}$ for a prosumer j (Equation (16)), while w_k is the weight of the typical day k , i.e., the number of days in the year represented by this typical day.

2.2. Social Welfare and Nash Bargaining Optimization Approaches

The cooperative optimization model of the EC (Section 2.1) is solved using a “Social Welfare” (SW) optimization approach or a “Nash Bargaining” (NB) optimization approach, which differ in the objective function considered. The aim of the SW optimization is

to maximize the total economic benefit of the system, which is the sum of the annual incremental economic benefits of the EC members:

$$u^{incr,tot,SW} = \sum_i u_i^{incr} \quad (18)$$

The cooperative optimization model according to SW is concave (a maximization problem is considered) and linear, thereby having a limited computational complexity. However, the optimal value of the objective function according to SW (i.e., the maximum value of $u^{incr,tot,SW}$ in Equation (18)) could be found for very different combinations of values of u_i^{incr} , potentially leading to the maximum total economic benefit of the system, but with a very unbalanced, and so certainly unfair, distribution of the annual incremental economic benefits among users (i.e., the optimal values of u_i^{incr}). To avoid this issue, cooperative allocation criteria such as Shapley value and Nucleolus (which are the focus of next Section 2.3) are applied to the SW optimal solution to re-distribute the maximum total economic benefit among users as fairly as possible.

In an NB approach, users find an agreement by cooperating within the EC to simultaneously maximize their annual incremental economic benefits and the total benefit of the system. The objective function to be maximized is the Nash product:

$$\prod_i u_i^{incr} \quad (19)$$

According to cooperative game theory, the NB solution has some desirable properties, namely “individual rationality” (i.e., each user has an incremental economic benefit obtained from cooperation within the EC compared to the disagreement case, see constraints (12) and (14) in Section 2.1), “efficiency” (i.e., the sum of the incremental economic benefits of users is equal to the total incremental benefit of the EC) and “Pareto optimality” (i.e., a user cannot increase its optimal incremental benefit without decreasing the incremental benefits of other users). However, the main issue with the NB is that the objective function in Equation (19) is non-convex and non-linear, which complicates the search for the global optimal solution for both the system and the users. Nevertheless, assuming that the incremental economic benefits are non-negative (as stated in constraints (12) and (14), Section 2.1), the NB solution can be found by an equivalent concave optimization problem [42,43,54]. As explicitly pointed out by Yaacoub and Dawy [55] and Yaiche et al. [56], it follows that

$$\max \prod_i u_i^{incr} \equiv \max \log \left(\prod_i u_i^{incr} \right) \equiv \max \sum_i \log(u_i^{incr}) \quad (20)$$

Thus, the NB optimization problem is equivalent to a concave optimization problem with an objective function given by the sum of logarithmic functions of u_i^{incr} . The solution of this equivalent optimization problem ensures “proportional fairness” in allocation problems according to Kelly et al. [57,58]. In the objective function $\sum_i \log(u_i^{incr})$, the higher the incremental economic benefit u_i^{incr} , the lower the increase in $\log(u_i^{incr})$. Compared to the SW approach (where the objective function is a sum of u_i^{incr}), in the equivalent problem to NB, users with low (or, in any case, not the highest) values of u_i^{incr} are given a higher weight because increasing a lower value of u_i^{incr} leads to a higher increase in $\log(u_i^{incr})$. Thus, the NB approach leads to a “proportional fair” solution, where users with low (or, in any case, not the highest) values of u_i^{incr} are assigned higher optimal values of u_i^{incr} compared to the SW approach. Moreover, in the equivalent optimization problem to NB, the individual terms $\log(u_i^{incr})$ are strictly concave (indeed, u_i^{incr} depend linearly on the P2G and P2P energy flows, see Equations (9)–(17) in Section 2.1), and consequently, the objective function $\sum_i \log(u_i^{incr})$ is also strictly concave [59], ensuring that the proportional fair NB solution is unique. The strict concavity of both the individual terms $\log(u_i^{incr})$ and the objective function $\sum_i \log(u_i^{incr})$ results in an optimal global solution for both the individual users and

the system as a whole, thus avoiding the case of finding an optimal solution for the system that is not optimal for the individual users (as could happen with SW).

Since the NB solution is the optimal trade-off between the maximum individual benefits and the maximum total benefit of the system (i.e., it is Pareto optimal), the sum of the optimal values of the annual incremental economic benefits (i.e., $u^{incr,tot,NB}$ in Equation (21)) could be lower than the SW approach (i.e., $u^{incr,tot,SW}$ in Equation (18)). This discrepancy is called “Price of Fairness” (POF) [60] and represents the amount of total economic benefits that are lost under the proportional fair NB solution compared to the SW solution. The POF is calculated as follows:

$$POF = \frac{u^{incr,tot,SW} - u^{incr,tot,NB}}{u^{incr,tot,SW}} \cdot 100 [\%] \quad (21)$$

where $u^{incr,tot,SW}$ is the optimal value of the objective function according to the SW approach (Equation (18)) and $u^{incr,tot,NB}$ is the sum of the optimal values of the annual incremental economic benefits found by the NB approach (Equation (19)).

2.3. Shapley Value and Nucleolus Allocation Criteria

The SW optimization of an EC could lead to an unfair distribution of the economic benefits among its members, as this optimization only aims at maximizing the total economic benefit of the system. This paper addresses such a problem by applying two allocation criteria based on cooperative game models, i.e., the Shapley value and the Nucleolus, after the SW optimization.

A cooperative game model is defined by N players forming the “grand coalition” (here the EC), which includes 2^N coalitions comprising the grand coalition itself and the empty coalition (i.e., the coalition without players) [32]. Each coalition S is associated with a “value” function, which represents the benefit of cooperation for the players. In this paper, the value of a coalition S is obtained as the optimal value of the objective function of a SW problem (Equation (18) in Section 2.2) that considers only the users in coalition S . Moreover, the value of the grand coalition corresponds to the sum of the optimal annual incremental economic benefits of all EC members found by SW.

According to the Shapley value criterion, the economic benefit x_i allocated to a member i of the EC is calculated as follows:

$$x_i = \sum_{S \subseteq \{1, \dots, N\}, i \in S} \frac{(|S| - 1)!(N - |S|)!}{N!} (v(S) - v(S \setminus \{i\})) \quad (22)$$

where x_i corresponds to the re-distributed annual incremental economic benefit (cost saving or profit increase, with respect to the disagreement case, that is different compared to that found by the SW optimization only), $|S|$ is the size of the coalition S (i.e., the number of members in the coalition) and $(v(S) - v(S \setminus \{i\}))$ is the contribution of the member i to the value $v(S)$ of coalition S , i.e., the difference between the value $v(S)$ with member i and the value $v(S \setminus \{i\})$ without member i . The Shapley value assigns a cost saving to each member (or profit increase) that represents its weighted average marginal contribution to the cost saving (or profit increase) of each coalition it participates in within the grand coalition. In other words, the Shapley value criterion satisfies the notion of “individual fairness” [31] in the sense that it succeeds in distributing the optimal incremental economic benefit of the EC by weighting the individual contributions of members to the total benefits of the EC and each coalition (i.e., $v(S)$ in Equation (22)) they participate in within the EC [35].

The Nucleolus criterion [32] is based on an iterative optimization procedure to minimize the “excess” of all coalitions, i.e., the “dissatisfaction” of the members of these coalitions if their benefits allocated within the grand coalition are lower than those they could obtain in independent coalitions outside of the grand coalition. The excess of a

coalition S , which is the difference between the value of the independent coalition and the sum of the benefits allocated to its members within the EC [61], is calculated as follows:

$$e_S = v(S) - \sum_{i \in S} x_i \quad (23)$$

The optimization procedure of Nucleolus is based on linear programming problems solved sequentially in different iterations [61], where the decision variables are the economic benefits x_i allocated to the EC members (i.e., the re-distributed annual incremental economic benefits) and the maximum excess among all coalitions. In this optimization, the equality between the value of the grand coalition (i.e., the optimal value of the objective function according to the SW approach, Equation (18) in Section 2.2) and the sum of the allocated economic benefits x_i is set to guarantee the “efficiency” property. Moreover, the excess of each coalition (Equation (23)) is set to be lower than or equal to the maximum excess. In the first iteration, the Nucleolus criterion identifies the coalitions with maximum excess and minimizes it. In the subsequent iterations, with the maximum excess fixed as a parameter, the excess of the other coalitions is minimized until all coalitions have a negative excess. This means that the sum of the benefits allocated to the users of each coalition within the EC is always greater than the economic benefits that each coalition could obtain if it operated independently. Thus, the EC is stable with no coalition of users willing to leave the EC, which is consistent with the notion of “collective fairness” [31] in the distribution of the incremental economic benefits.

Although the Shapley value weights the individual contributions of users to the values of all coalitions (according to “individual fairness”), some coalitions could have positive excesses, contributing to the instability of the EC. On the other hand, the Nucleolus always guarantees stability (according to “collective fairness”), but the higher contributions of some users to the benefits of the EC and its coalitions could be less rewarded.

2.4. Input Data

The PV plants of the agricultural (Agr) and residential (Res2) prosumers (Figure 1) have a surface area of 60 m² and 40 m² respectively, which fully cover their peak demand, considering about 6 m² of area per kW [11]. The grid capacity P^{max} (constraints (4) and (5) in Section 2.1) is assumed to be 70 kW [62].

The daily timeseries of solar irradiance refer to the location of Padova (Italy) and are taken from the historical data (2005–2020) of the PVGIS database [63]. The daily timeseries of electricity demands for the commercial (Com) consumer, residential (Res1) consumer, Agr and Res2 prosumers are taken from [64] and appropriately scaled to ensure heterogenous profiles and, at the same time, a good match with the PV generation profiles.

The daily timeseries of the P2G electricity prices are based on a historical dataset (2005–2020) of the day-ahead market price in Italy provided by the Italian market operator (called “GME”) [65]. The P2G sale price is equal to the day-ahead market price, while the P2G purchase price is calculated as the P2G sale price plus a grid tariff. According to the mid-market pricing scheme [47], the mid-market rate price is defined hourly as the average of the P2G purchase and sale prices [47]. The P2P purchase and sale prices are then defined hourly as the average between the mid-market rate price and the P2G purchase and sale prices, respectively, thus ensuring attractive P2P trading prices for both consumers and prosumers within the EC. It is worth noting that there are more sophisticated P2P pricing schemes, which are based on the dynamic equilibrium between energy demand and supply within the EC [13,20]. However, these schemes, similarly to the mid-market one, constrain the P2P prices to be within the interval defined by the P2G purchase and sale prices, where this interval depends on the grid tariff. Hence, it is predicted that the specific P2P pricing scheme will have a limited impact on the different fair allocations of the economic benefits of an EC. On the contrary, since the P2P prices depend on the P2G prices and thus on the grid tariff, the value of the grid tariff is fixed at 0.1 €/kWh [14] and 0.015 €/kWh [64] in

scenario 1 and scenario 2, respectively, considered in Section 3, in order to assess the impact of the extreme values of this parameter on the allocation of the economic benefits to users.

Figure 2a,b and Figure 3 show the 8 typical days of global solar irradiance, user electricity demands and P2G sale prices representative of a typical year, respectively. These typical days are obtained by applying the K-means clustering algorithm to the considered historical dataset covering 16 years, and the weights of the typical days are then adjusted to comply with the annual time horizon of the operation optimization (Section 2.1). Figure 4a,b show the daily profiles of the P2G and P2P purchase and sale prices for one of the 8 typical days, in scenario 1 and scenario 2 of the grid tariff, respectively.

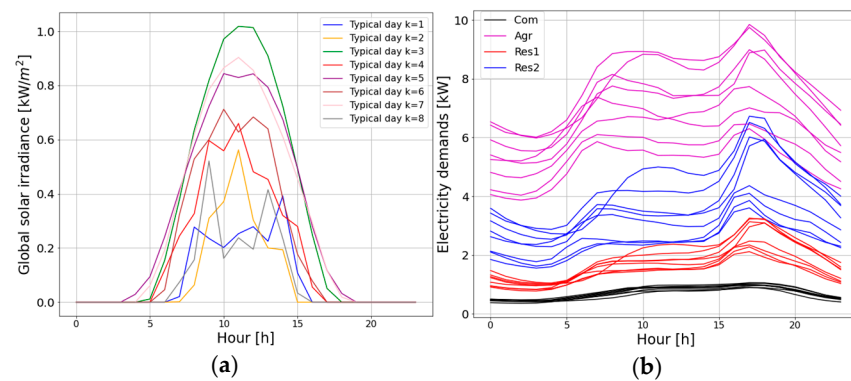


Figure 2. The figure shows 8 typical days of (a) global solar irradiance in Padova (Italy) and (b) electricity demands of the commercial (Com) and residential (Res1) consumers as well as the agricultural (Agr) and residential (Res2) prosumers.

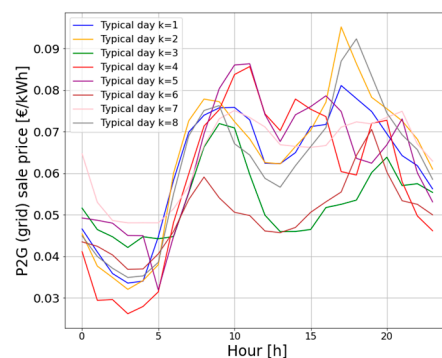


Figure 3. The figure shows 8 typical days of the Peer-to-Grid (P2G) sale price.

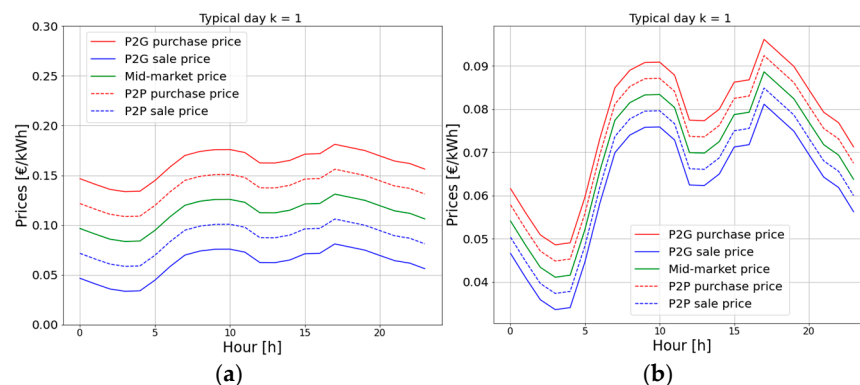


Figure 4. Peer-to-Grid (P2G) and Peer-to-Peer (P2P) purchase and sale prices for one typical day, considering (a) scenario 1 (grid tariff of 0.1 €/kWh) and (b) scenario 2 (grid tariff of 0.015 €/kWh).

3. Results and Discussion

The cooperative optimization model of the EC (Section 2.1) is developed in Pyomo [66,67] and solved according to the “Social Welfare” (SW) and “Nash Bargaining” (NB) optimization approaches using Gurobi software (version 11.0.1) [68] with a maximum optimality gap of 1×10^{-4} . The computer utilized is an Intel(R), Core (TM) i5-8350U with 1.70 GHz and 16 GB of RAM. The computational time to solve the cooperative model of the EC and obtain the fair economic allocation is a few seconds with the NB approach (considering the equivalent concave optimization problem, Section 2.2) and less than half a minute with the SW approach and the subsequent application of the Shapley value or Nucleolus criterion (Section 2.3).

Section 3.1 shows the analyses of the results obtained by solving the cooperative model of the EC with the SW and NB optimization approaches, highlighting the differences in the optimal incremental economic benefits of the users (i.e., their annual cost savings). Section 3.2 presents the results of applying the Shapley value and Nucleolus criteria to the SW optimal solution, which lead to different fair re-distributions of the optimal incremental economic benefit of the EC. Section 3.3 compares the fair distribution of the optimal total incremental economic benefit obtained by the NB optimization approach with the fair re-distributions of the optimal total incremental economic benefit of SW using the Shapley value and Nucleolus criteria. As the calculation of the Peer-to-Peer (P2P) prices depends on the Peer-to-Grid (P2G) prices and thus on the grid tariff, the following results are obtained considering two different values of the grid tariff, i.e., 0.1 €/kWh and 0.015 €/kWh, hereafter referred to as scenario 1 and scenario 2, respectively.

3.1. Social Welfare and Nash Bargaining

Table 1a,b show the optimal annual costs (as economic benefits with negative values) found in the “disagreement” case (Section 2.1) and in the cooperative case by solving the SW and NB optimizations, considering a grid tariff of 0.1 €/kWh in scenario 1 and 0.015 €/kWh in scenario 2, respectively. As expected, all of the optimal costs are higher in scenario 1 associated with a higher grid tariff. In both scenarios 1 and 2, the “disagreement” case is the least economically convenient way of operation, resulting in higher annual costs compared to the SW and NB solutions, because users operate independently without cooperation in the EC and thus do not benefit from energy sharing at the P2P prices (which are more economically convenient than the P2G ones). Moreover, the optimal total cost found by NB, which maximizes both the individual economic benefits of users and the total benefit of the EC, is only slightly higher than that of SW, which considers only the total benefit and not the individual benefits (i.e., € −11,953.54 with NB and € −11,949.94 with SW in scenario 1, and € −5429.86 with NB and € −5423.79 with SW in scenario 2).

Figure 5a,b show, on the left axis, the optimal annual cost savings (i.e., the optimal incremental benefits u_i^{incr} compared to the “disagreement” case, Equation (17) in Section 2.1) obtained by the SW (gray bars, before the re-distribution using Shapley value or Nucleolus) and NB (blue bars) optimization approaches and, on the right axis, the relative difference in the annual cost savings between SW and NB (black dots), considering the SW as reference, in scenarios 1 and 2, respectively. The optimal annual cost savings correspond to the differences between the optimal annual costs (in absolute value, Table 1a,b) of the “disagreement” case and those found with SW or NB. The SW approach leads to the lowest annual cost savings for the commercial (Com) consumer (i.e., around € 24 and € 9 in scenarios 1 and 2, respectively), while agricultural (Agr) and residential (Res2) prosumers achieve higher annual cost savings. Thus, the SW approach rewards prosumers more than consumers, neglecting that consumers also play an important role in realizing the P2P energy sharing within the EC, which results in the incremental economic benefits for all users and the whole EC compared to the “disagreement” case. Hence, the distribution of annual cost savings by SW (Equation (18), Section 2.2), which does not optimize the individual annual incremental economic benefits, without the re-distribution using Shapley value or Nucleolus, may be perceived as unfair by consumers. On the contrary, looking at the black dots in

Figure 5a (Figure 5b), the NB approach (Equation (19), Section 2.2) increases the optimal annual cost savings of Com by 39% (49%) in scenario 1 (2) and of Agr by 17% in scenario 1 (by 9% for Res2 in scenario 2) compared to SW, thus rewarding both the consumer with the lowest electricity demand (Com) and the prosumers who contribute to the P2P energy sharing (Agr and Res2). This result is consistent with the “proportional fair” NB solution (Equation (20), Section 2.2), where users with low values of the annual incremental benefits (e.g., Com) or users with not the highest values of the annual incremental benefits according to SW (e.g., Agr and Res2 in scenarios 1 and 2, respectively) can achieve higher benefits compared to SW. On the other hand, users with the highest values of the annual incremental benefits according to SW (e.g., Res2 and Agr in scenarios 1 and 2, respectively) have the highest decrease in their benefits by moving from the SW solution to the NB one (i.e., by 18% and 24% for Res2 and Agr in scenarios 1 and 2, respectively). Furthermore, comparing Figure 5a,b, according to the NB solution, the optimal annual cost saving of Agr increases/decreases by 17%/24% in scenario 1/2 (on the contrary, the optimal annual cost saving of Res2 decreases/increases by 18%/9% in scenario 1/2) compared to SW. The reason for this is that the NB solution, which seeks to maximize both individual and total benefits, is more balanced than the SW solution, leading to smaller differences in the annual cost savings between the different users. For example, moving from the SW solution to the NB solution in scenario 2 (Figure 5b), the annual cost savings of Agr and Res2 change from € 52 and € 30, respectively, to € 40 and € 33, respectively. Thus, NB limits the highest annual cost saving of Agr (user with the largest PV plant), which is derived from selling P2G and P2P energy (which is very profitable in scenario 2 as the P2G and P2P sale prices are close to the P2G and P2P purchase prices).

Table 1. Annual costs of users obtained in the “disagreement” case and by solving the cooperative model of the EC according to the Social Welfare and Nash Bargaining optimization approaches, considering (a) scenario 1 (grid tariff of 0.1 €/kWh) and (b) scenario 2 (grid tariff of 0.015 €/kWh).

(a)			
Annual Costs [€], Scenario 1 with Grid Tariff Equal to 0.1 €/kWh			
Member	Disagreement Case	Nash Bargaining	Social Welfare
Com	−1000.52	−967.18	−976.65
Res1	−2285.05	−2241.50	−2235.77
Agr	−6153.55	−6073.26	−6084.82
Res2	−2751.90	−2671.60	−2652.69
Total	−12,191.02	−11,953.54	−11,949.94
(b)			
Annual Costs [€], Scenario 2 with Grid Tariff Equal to 0.015 €/kWh			
Member	Disagreement Case	Nash Bargaining	Social Welfare
Com	−476.17	−463.40	−467.61
Res1	−1085.24	−1066.80	−1066.33
Agr	−2794.09	−2754.18	−2741.61
Res2	−1178.48	−1145.49	−1148.24
Total	−5533.98	−5429.86	−5423.79

Looking again at Figure 5a,b, the dashed orange circles highlight the Price of Fairness (POF) (Equation (21), Section 2.2), which is the relative decrease in the optimal annual cost savings of the EC by moving from the SW solution to the proportional fair NB solution. In scenario 1, the POF is 1.35% (in absolute value) with optimal annual cost savings of the EC around € 241 and € 238 according to SW and NB, respectively, while in scenario 2, the POF is 5.51% (in absolute value) with optimal total annual cost savings around € 110 and € 104 according to SW and NB, respectively.

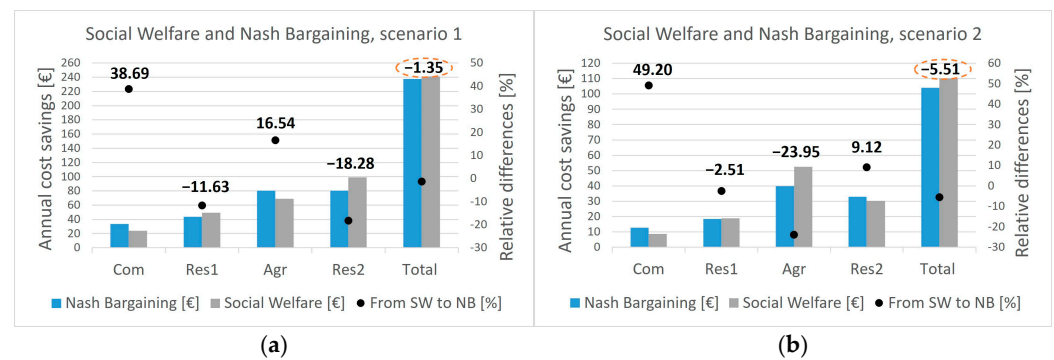


Figure 5. Optimal annual cost savings [€] obtained by using the Social Welfare (SW) and Nash Bargaining (NB) optimization approaches (left axis) and relative difference [%] in the annual cost savings between SW and NB considering SW as reference (right axis), in (a) scenario 1 (grid tariff of 0.1 €/kWh) and (b) scenario 2 (grid tariff of 0.015 €/kWh). The dashed orange circles highlight the Price of Fairness, i.e., the reduction in the total annual cost savings by moving from the SW solution to the NB solution.

Considering the NB solution and the 8 typical days of the optimization, Figure 6a,b show the daily profiles of the optimal P2G power of the users in scenarios 1 and 2, respectively, while Figure 7a,b show the daily profiles of the optimal P2P power exchanged between the users in scenarios 1 and 2, respectively. Positive and negative values of P2G and P2P power refer to power purchased and sold, respectively. It is interesting to note that Agr and Res2 prosumers, in scenarios 1 and 2, respectively, make more use of the P2P energy sharing (purple line in Figure 7a for Agr and blue line in Figure 7b for Res2) to sell electricity to the consumers (Com and Res1), rather than selling electricity to the grid (purple line in Figure 6a for Agr and blue line in Figure 6b for Res2). The reason for this is that the NB solution, which satisfies the “proportional fairness”, rewards the P2P energy sharing between users more than the P2G energy exchanged, also resulting in an increase in the annual cost savings of Agr and Res2 by about 17% and 9% with NB compared to SW in scenarios 1 and 2, respectively (see Figure 5a,b).

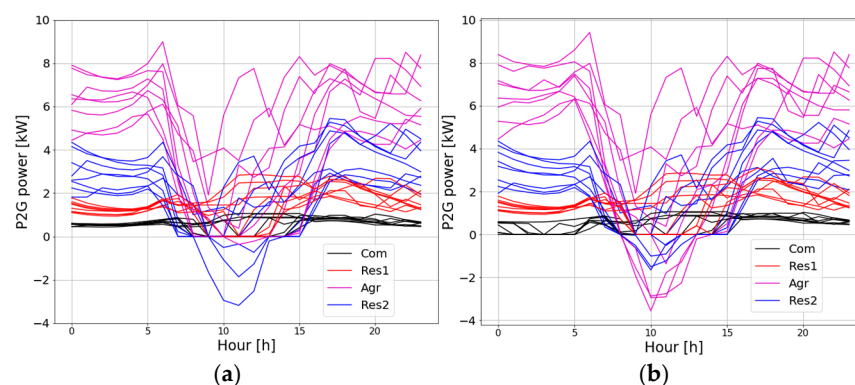


Figure 6. Optimal P2G power of users in the 8 typical days considered by the NB optimization, in (a) scenario 1 (grid tariff of 0.1 €/kWh) and (b) scenario 2 (grid tariff of 0.015 €/kWh).

Considering again the NB solution and the 8 typical days of the optimization in scenario 1, Figure 8 shows the input electricity demands of the users ((a) Com, (b) Res1, (c) Agr and (d) Res2)) and their optimal shifted electricity demands due to the PBDR (solid and dashed lines respectively) and the daily profiles of the PV production ((c) Agr and (d) Res2)). The application of PBDR leads to an increase in the demand in the middle of the day for all users, thus improving the match with the PV production. This also allows for the P2P energy sharing between consumers (Com and Res1) and prosumers (Agr and Res2), as already highlighted in Figure 7. However, it is found that the optimal shifted electricity

demands do not change significantly between the NB and SW solutions, as well as between scenarios 1 and 2, indicating their low impact on the distribution of the optimal annual cost savings among the EC members. On the other hand, the P2G and P2P energy flows directly affect the individual economic benefits of users (Equations (9) and (10), Section 2.1) and, therefore, the distribution of the optimal annual cost savings of the EC.

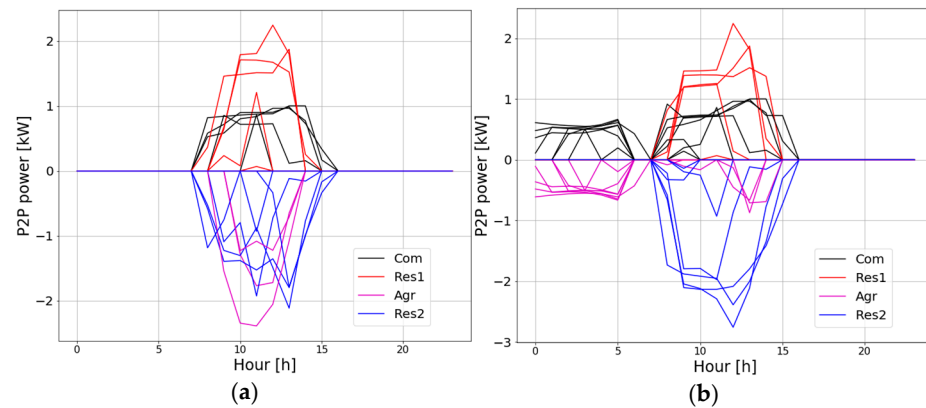


Figure 7. Optimal P2P power between users in the 8 typical days considered by the NB optimization, in (a) scenario 1 (grid tariff of 0.1 €/kWh) and (b) scenario 2 (grid tariff of 0.015 €/kWh).

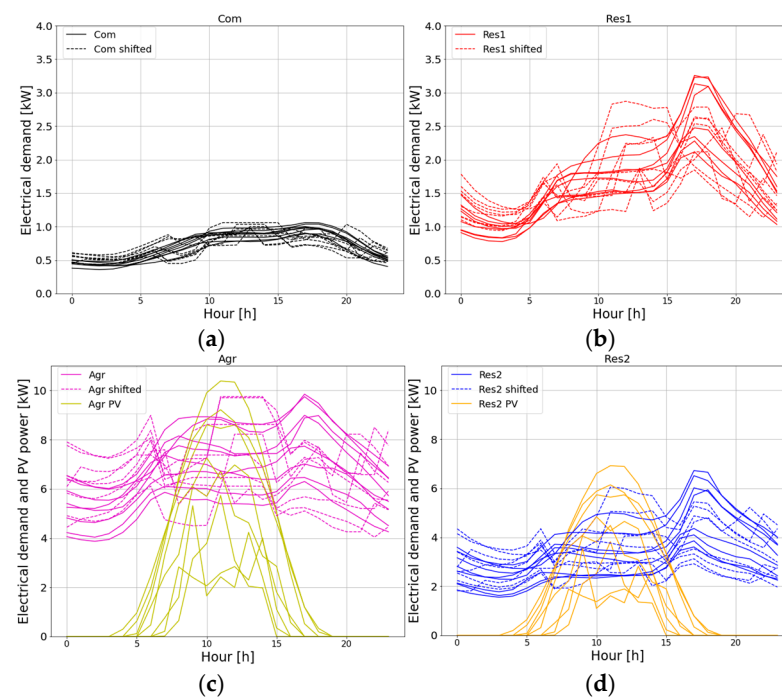


Figure 8. Input electricity demands and optimal shifted electricity demands due to the PBDR (solid and dashed lines, respectively), and PV generation profiles, in the 8 typical days considered by the NB optimization in scenario 1 (grid tariff of 0.1 €/kWh) for the (a) Com, (b) Res1, (c) Agr and (d) Res2 users.

Finally, the relationship between the value of the grid tariff, which distinguishes the two scenarios considered, and the value of the POF should be discussed. As shown in Figure 5a (Figure 5b), the lowest (highest) POF of 1.35% (5.51%) is obtained with the highest (lowest) value of the grid tariff of 0.1 (0.015) €/kWh in scenario 1 (scenario 2). The higher the grid tariff (scenario 1), the higher the differences between the P2G and P2P prices (Figure 4a), also resulting in higher P2G and P2P purchase prices. Hence, P2P energy sharing is mainly profitable only in the presence of high solar PV generation (i.e., in the middle of the day as in Figure 7a), which is the most likely outcome for both the SW and NB optimizations. As a result, the NB and SW solutions become closer, leading to a low

value of the POF. On the other hand, the lower the grid tariff (scenario 2), the smaller the differences between the P2G and P2P prices (Figure 4b). In this scenario, users could also exchange P2P energy during hours of the day when solar PV generation is very low (as in Figure 7b), leading to an NB solution that deviates more from the SW solution, and thus, a higher POF, compared to scenario 1. Note that in both scenarios, the optimal SW solution is based on using the P2P energy sharing only in the middle of the day to exploit the total PV self-consumption and, thus, only maximize the total economic benefit of the EC.

3.2. Shapley Value and Nucleolus

Before applying the Shapley value and Nucleolus criteria (Section 2.3) to re-distribute the total incremental economic benefit among the EC members, the value of each coalition of users is found by solving the cooperative optimization model, using the SW approach, for each coalition. As an example, Table 2 shows the coalition values representing the optimal annual cost savings of coalitions in scenario 1 (grid tariff of 0.1 €/kWh), where users 1, 2, 3 and 4 correspond to users Com, Res1, Agr and Res2, respectively. It is worth noting that coalitions {1}, {2}, {3} and {4} represent independent users that operate as in the “disagreement” case, and therefore, their annual cost savings are zero (Equations (11)–(14) in Section 2.1), since there is no possibility of sharing P2P energy with other EC members. The annual cost saving of coalition {1,2,3,4} is that of the EC (i.e., € 241 in scenario 1, Figure 5a).

Table 2. Values of user coalitions, i.e., optimal values of the annual cost savings of user coalitions, in scenario 1 (grid tariff of 0.1 €/kWh).

Annual Cost Savings of Coalitions [€], Scenario 1 with Grid Tariff Equal to 0.1 €/kWh				
{1}: 0	{1,2}: 18.46	{1,2,3}: 124.65	{1,2,3,4}: 241.08	{1,2,4}: 177.6
{1,3}: 87.17	{1,3,4}: 169.95	{1,4}: 113.68	{2}: 0	{2,3}: 110.04
{2,3,4}: 208.45	{2,4}: 153.65	{3}: 0	{3,4}: 109.02	{4}: 0

Figure 9a,b show, on the left axis, the re-distribution of the optimal annual cost savings of the EC by Shapley value (blue bars) and Nucleolus (gray bars) and, on the right axis, the relative differences in the re-distributed annual cost savings by Shapley value (orange dots) and Nucleolus (yellow dots) compared to those obtained by the base SW optimization only (i.e., without using the Shapley value and Nucleolus after the SW optimization, Figure 5a,b), in scenarios 1 and 2, respectively.

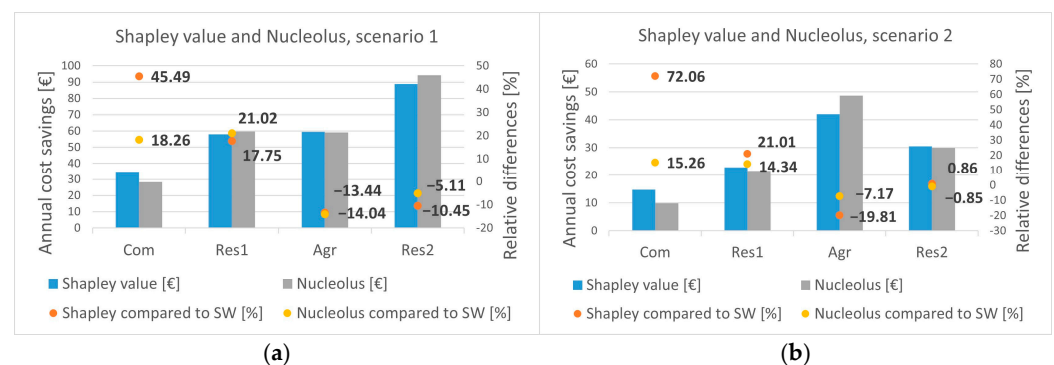


Figure 9. Allocation of the optimal annual cost savings [€] by Shapley value and Nucleolus (left axis) and relative differences [%] in the re-distributed annual cost savings compared to those obtained with only the Social Welfare (SW) approach (right axis), in (a) scenario 1 (grid tariff of 0.1 €/kWh) and (b) scenario 2 (grid tariff of 0.015 €/kWh).

In both scenarios, Shapley value increases the annual cost savings of consumers Com and Res1 (e.g., in scenario 2, by about 72% and 21%, respectively) and generally reduces the annual cost savings of the prosumers Agr and Res2 (e.g., in scenario 2, by about 20%

for Agr) compared to the base SW optimization. Thus, compared to the base SW, the Shapley value rewards the low electricity consumption of consumers more than the solar PV generation of prosumers, who have a high share in the optimal annual cost of the EC (e.g., Agr contributes 50% of the total annual costs, see Table 1a,b). This result is a consequence of the “individual fairness” ensured by the Shapley value criterion, which re-distributes the total annual cost savings of SW by weighting the individual contribution of each EC member (Equation (22) in Section 2.3) to the annual cost savings of each coalition and the EC (i.e., the values shown in Table 2).

Similar to the Shapley value, Nucleolus also increases/decreases the annual cost savings of consumers/prosumers compared to those of the base SW (e.g., in scenario 2, the annual cost savings of Com and Res1 increase by about 15% and 14%, while those of Agr and Res2 decrease by about 7% and 0.9%), but there are smaller relative variations compared to the allocation by Shapley value (Figure 9). This difference can be explained by calculating the excess of coalitions in the Shapley value allocation. The excess of a coalition (Equation (23) in Section 2.3) is calculated as the difference between its optimal annual cost saving (i.e., the coalition value, Table 2) and the sum of the annual cost savings re-distributed to its members according to the Shapley value criterion (Figure 9). Figure 10a,b show the excess of all coalitions within the EC in scenarios 1 and 2, respectively. In scenario 1, coalitions $\{2,3,4\}$ and $\{2,4\}$ have positive excesses (i.e., € 2.1 and € 6.79, respectively), meaning that these coalitions could achieve higher annual cost savings if they operated independently and not within the whole EC. Given the high grid tariff in scenario 1, P2P energy sharing between Res1 (user 2) and Res2 (user 4) is profitable when they operate in smaller coalitions than the whole EC, i.e., in the unstable coalitions found ($\{2,3,4\}$ and $\{2,4\}$). Therefore, these coalitions and thus Res1 and Res2 might want to leave the EC, thus leading to instability. This problem is neglected in the Shapley value allocation, but is addressed by Nucleolus, which minimizes the excess of all coalitions through an iterative optimization procedure. After applying this optimization (Section 2.3), all excesses become negative (e.g., −4.4 and −0.13 for coalitions $\{2,3,4\}$ and $\{2,4\}$, respectively). To address the problem of instability raised by coalitions $\{2,3,4\}$ and $\{2,4\}$ in scenario 1 with Shapley value, Nucleolus assigns higher annual cost savings to users Res1 and Res2 than the Shapley value (see Figure 9a), i.e., the annual cost savings of Res1 and Res2 increase by 21% (18% with Shapley value) and decrease by 5% (10% with Shapley value), respectively, compared to SW.

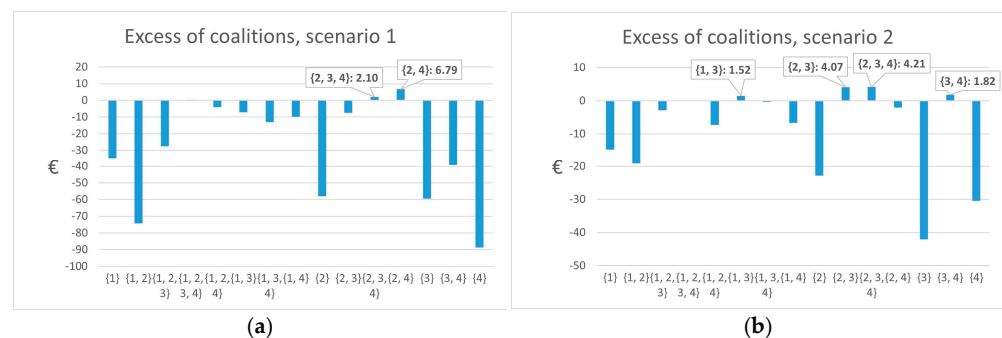


Figure 10. Excess of each coalition, given the allocation of the annual cost savings by Shapley value, in (a) scenario 1 (grid tariff of 0.1 €/kWh) and (b) scenario 2 (grid tariff of 0.015 €/kWh). Negative and positive excess values represent coalitions willing to stay in the EC and willing to leave the EC, respectively.

At the same time, Nucleolus reduces the annual cost savings of Com and Agr (e.g., the increase in the annual cost saving of Com compared to SW changes from 45% with Shapley value to 18% with Nucleolus). Looking at Figure 10b for scenario 2, coalitions $\{1,3\}$, $\{2,3\}$, $\{2,3,4\}$ and $\{3,4\}$ show positive excesses. The common user in all these coalitions is the Agr prosumer (user 3), who is responsible for the instability of the EC (i.e., given the low grid tariff in scenario 2, P2G and P2P sale prices are close to the P2G and P2P purchase prices,

and selling electricity is profitable for Agr that owns the largest PV plant). Consequently (see Figure 9b), Nucleolus increases the annual cost saving of Agr (i.e., the decrease in the annual cost saving compared to SW moves from 20% with Shapley value to 7% with Nucleolus) and decreases the annual cost savings of other users (e.g., the increase in the annual cost saving of Com compared to SW changes from 72% with Shapley value to 15% with Nucleolus). The results obtained show that the Nucleolus criterion leads to a more homogeneous re-distribution of the annual cost savings across users than the Shapley value, with relative variations in the annual cost savings over SW being more similar across the different users (Figure 9). Nucleolus ensures “collective fairness” among the EC members, i.e., it prevents user coalitions from leaving the EC, thus maintaining stability. On the other hand, Nucleolus could make an allocation that is less “individually fair” compared to Shapley value.

3.3. Critical Remarks

Based on the results in Sections 3.1 and 3.2, the following list summarizes and compares the main findings on the different fair allocations of the incremental economic benefits of an EC (i.e., the annual cost savings compared to a “disagreement case”), considering both scenarios of the grid tariff (i.e., 0.1 €/kWh in scenario 1 and 0.015 €/kWh in scenario 2).

- The cooperative operation of users, with the possibility of exchanging energy with the electric distribution grid (i.e., realizing P2G energy flows) and sharing energy within the EC (i.e., realizing P2P energy flows), is always profitable compared to the “disagreement” case, where each user operates individually by exchanging energy only with the grid (Table 1).
- The NB optimization approach provides a distribution of the optimal annual cost savings that satisfies the notion of “proportional fairness”, rewarding both the consumer with the lowest electricity demand (Com) and the PV prosumers (Agr and Res2) that contribute to the P2P energy sharing (Figures 5–7). Compared to the base optimal SW solution (i.e., without re-distribution using Shapley value or Nucleolus), the optimal NB one allocates 39% (49%) and 17% (9%) higher annual cost savings to consumer Com and prosumer Agr (Com and Res2) in scenario 1 (2), respectively, with a POF (i.e., the reduction in the optimal total cost savings of the EC compared to those of the SW solution) of 1.4% (5.5%), in exchange for a distribution of the optimal annual cost savings that could be perceived as fair by both consumers and prosumers of the EC. The low POF sheds light on the low “price” to be paid (i.e., a small part of the total annual cost savings of the EC) to switch from a SW solution to an NB solution, which, being “proportional fair”, can satisfy both consumers with low electricity demands and prosumers contributing to the P2P energy sharing (and not only prosumers, as in the base SW solution, which is also unfair since prosumers have higher electricity demands and higher shares in the annual costs of the EC than consumers, Table 1). Thus, the main difference between the SW and NB solutions does not lie in the *optimal total annual cost savings* of the EC, but in how the *optimal individual annual cost savings* are distributed among the different EC members.
- The Shapley value criterion provides a re-distribution of the optimal total annual cost savings, found by SW, which is consistent with the notion of “individual fairness”, rewarding the users with the lowest electricity consumption more (Com and Res1 consumers) because the electricity demand of some users (Agr and Res2 prosumers) strongly influences the annual operational cost of the EC. Compared to the base optimal SW solution, Shapley value allocates 45% (72%) and 18% (21%) higher annual cost savings to Com and Res1 consumers with the lowest electricity demand in scenario 1 (2) (Figure 9). However, while Shapley value allocation might be appreciated by the most virtuous EC members with low electricity consumption, it could lead to unstable coalitions willing to leave the EC (see Figure 10).
- The Nucleolus criterion, also applied to re-distribute the optimal total annual cost savings of the EC found by SW, provides an alternative solution to the Shapley value

allocation. Similar to the Shapley value, the Nucleolus re-distributes higher annual cost savings to Com and Res1 consumers compared to the base optimal SW solution (i.e., 18% and 21% higher in scenario 1, 15% and 14% higher in scenario 2, respectively). Moreover, the Nucleolus allocation embodies the notion of “collective fairness”, which ensures that all user coalitions are satisfied with the allocation of the total annual cost savings, resulting in a stable EC without objective reasons for coalitions to leave the EC. To avoid the instability problem raised by some coalitions in the Shapley value allocation, Nucleolus increases the annual cost savings of Res1 and Res2 in scenario 1 and Agr in scenario 2 compared to Shapley value just enough to persuade them to stay in the EC. For example, in scenario 1, the annual cost savings of Res1 and Res2 increase by 21% (18% with Shapley value) and decrease by 5% (10% with Shapley value), respectively, compared to the base optimal SW solution. However, the homogeneous distribution of annual cost savings by Nucleolus could be perceived as less “individually fair” compared to that of the Shapley value.

In general, there is not a unique better method to fairly allocate the total economic benefits of an EC, as fairness is a subjective matter and different notions of fairness exist [25,31]. However, it seems that the NB optimization approach, which leads to a “proportional fair” solution, is the best compromise among the methods applied. Figure 11a,b compare the relative differences in the optimal individual annual cost savings obtained via the Shapley value and Nucleolus criteria and the NB optimization, with respect to the base SW optimization in scenarios 1 and 2, respectively. As highlighted by the dashed black circles, the NB solution improves the annual cost savings of both a consumer and a prosumer in each scenario compared to the base SW solution, thus achieving a better balance between the interests of both consumers and prosumers than the Shapley value and Nucleolus. In other words, while the Shapley value and Nucleolus criteria reward only the EC members with the lowest electricity consumption (i.e., the consumers in the analyzed EC, Figure 9), the NB approach rewards both the consumer with the lowest electricity consumption and the prosumers who promote the P2P energy sharing within the EC (see also the optimal annual cost savings of Com, Agr and Res2 in Figure 5, and the P2G and P2P energy flows of Agr and Res2 in Figures 6 and 7).

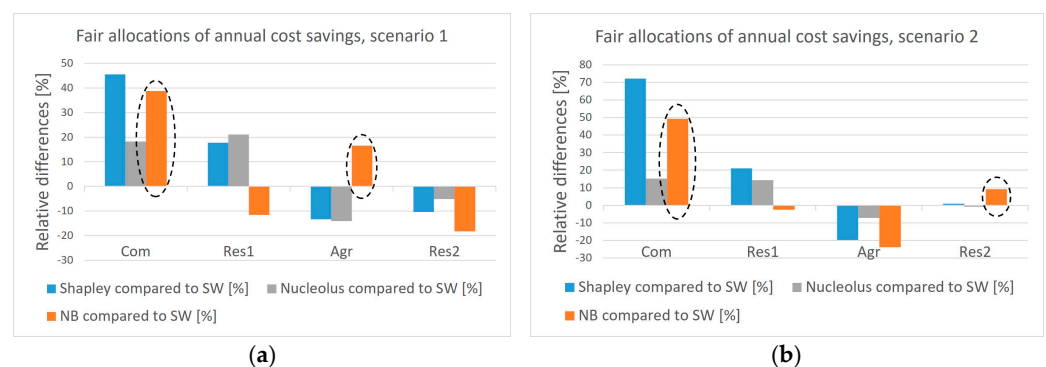


Figure 11. Relative differences [%] in the optimal annual cost savings of the users, allocated by the Shapley value and Nucleolus criteria and the Nash Bargaining (NB) optimization, compared to the base Social Welfare (SW) optimization, in (a) scenario 1 (grid tariff of 0.1 €/kWh) and (b) scenario 2 (grid tariff of 0.015 €/kWh). The dashed black circles show that NB allows to satisfy at least one consumer and one prosumer, contrary to Shapley value and Nucleolus.

Eventually, it should be emphasized that the proposed cooperative model is flexible, as it could be adapted to different EC settings characterized, for example, by a higher number of members than the one considered. In this case, the optimization approaches and cooperative allocation criteria used can still be applied, and the NB approach will have a clear computational advantage over the SW approach with Shapley value or Nucleolus criterion. In fact, both allocation criteria require the calculation of the value of each coalition

within the EC (Section 2.3), which means that the cooperative model is solved for each coalition. Considering a high number N of members (e.g., tens or hundreds) and, therefore, 2^N coalitions, the computational time to fairly re-distribute the total economic benefits of the EC by Shapley value and Nucleolus could increase drastically, thus requiring approximation methods [23,36].

4. Conclusions

The framework of Energy Communities (ECs) represents a significant shift in the way energy is produced, consumed and shared. ECs, which consist of aggregations of users that cooperate to achieve economic benefits through energy sharing, offer a promising alternative to the traditional model where users operate independently. This paper focuses on exploring different methodologies for optimizing and fairly allocating the economic benefits of an EC.

The concept of fairness in the cost/profit allocation for ECs is certainly subjective and, in general, there is no unambiguous definition of it. This paper aims to identify an allocation method that allows for a proper weighting of the main points of view of the different EC members, finding the optimal compromise between the interests of the community as a whole and those of each individual user, as well as between the interests of the energy consumers and prosumers. Although different notions of fairness have been presented in the literature, less attention has been paid to comparing different optimization approaches and cooperative allocation criteria that satisfy different notions of fairness. To fill this gap, the novel contribution of this work is to compare the solutions obtained by applying different optimization approaches and cooperative allocation criteria associated with the different notions of “individual”, “collective” and “proportional” fairness, with the aim of evaluating which method may be most appropriate to achieve an optimal compromise of the interests of all EC members. The used optimization approaches consider objective functions representing different combinations of the incremental economic benefits (i.e., cost savings and/or profit increases) obtained by users cooperating within the EC compared to a “disagreement” case in which they act individually and independently.

“Social Welfare” (SW) and “Nash Bargaining” (NB) optimization approaches are used to solve the cooperative model of an EC. The SW approach maximizes the total incremental economic benefit of the EC, while the NB approach maximizes both the total incremental benefit of the EC and the individual incremental benefit of each user, ensuring that the incremental benefits are distributed according to the notion of “proportional fairness”. In addition, since the SW optimization could lead to an unfair distribution of the total incremental benefit, the cooperative allocation criteria of Shapley value and Nucleolus are applied to re-distribute the optimal total incremental benefit among the EC members. The Shapley value allocates the optimal total incremental benefit of the EC according to the notion of “individual fairness”, i.e., by weighting the contribution of each member to the optimal benefits of each user coalition and the EC. On the other hand, the Nucleolus allocates the optimal total incremental benefit of the EC according to the notion of “collective fairness”, i.e., by ensuring that all user coalitions within the EC obtain higher benefits than by operating independently.

The optimization approaches and cooperative allocation criteria are applied to a case study of an EC consisting of four users, i.e., a commercial (Com) and a residential (Res1) consumer and an agricultural (Agr) and a residential (Res2) prosumer owning a solar PV plant, with flexible electricity demands that can be shifted according to a Price-Based Demand Response (PBDR). The EC members can use Peer-to-Grid (P2G) and Peer-to-Peer (P2P) mechanisms to exchange electricity with the grid and to share electricity among themselves, respectively. It is found that even shifting users’ electricity demand by PBDR could improve P2P energy sharing between consumers and prosumers (through the self-consumption of PV energy), PBDR has a limited impact on the different distributions of the optimal annual cost savings, which are most affected by the optimal P2G and P2P energy flows of users. Moreover, two different scenarios of the grid tariff (i.e., 0.1 €/kWh in

scenario 1 and 0.015 €/kWh in scenario 2), which affects the P2G and P2P purchase and sale prices, are considered to assess its impact on the different fair allocations obtained.

The following general guidelines emerge from the obtained results.

- The Shapley value and Nucleolus criteria can effectively reward users with low electricity consumption (i.e., Com and Res1 consumers in the considered case study). The Shapley value re-distributes 45–72% and 18–21%, respectively, while the Nucleolus re-distributes 15–18% and 14–21%, respectively, higher annual cost savings to Com and Res1 consumers compared to the base SW optimization. However, the Shapley value carries a risk of instability, as it could incentivize some coalitions to leave the EC due to perceived inequities associated with the lower benefits within the EC than in the case where the coalitions act independently. On the other hand, the Nucleolus criterion provides a more stable allocation, ensuring that all user coalitions are satisfied and want to stay in the EC, even though this allocation could be seen as less “individually fair” compared to the Shapley value (i.e., due to the smaller variations in the optimal annual cost savings with respect to the base SW solution).
- The NB approach rewards both the user with the lowest electricity consumption (i.e., Com) and PV prosumers (i.e., Agr and Res2) who contribute to the P2P energy sharing. Compared to the base SW optimization, NB distributes 39–49% higher annual cost savings to Com consumer and 9–17% higher savings to Agr and Res2 prosumers, albeit with an acceptable Price of Fairness (POF, i.e., the reduction in the optimal total benefit compared to the base SW solution) of 1.4% (scenario 1) and 5.5% (scenario 2).

Although the NB optimization approach, the Shapley value and Nucleolus criteria lead to different fair allocations of the optimal cost savings of an EC, the NB approach could generally be suggested as the most suited method for distributing the benefits of an EC composed of both consumers and prosumers, achieving a better balance between the economic interests of these different energy users than the Shapley value and Nucleolus. Future work will consider more complex ECs, characterized by a higher number of members (e.g., tens or more), with more energy units (e.g., storage units). Another interesting direction of future work will be to identify parameters that can quantitatively compare the different benefit distributions based on the “individual”, “collective” and “proportional” fairness.

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Nomenclature

Acronyms

Agr	Agricultural
Com	Commercial
EC	Energy Community
MILP	Mixed-Integer Linear Programming
NB	Nash Bargaining
PBDR	Price-Based Demand Response
POF	Price of Fairness
PV	Photovoltaic
P2G	Peer-to-Grid
P2P	Peer-to-Peer
RES	Renewable Energy Sources

Res	Residential
RTP	Real-Time Pricing
SW	Social Welfare
Symbols	
e_S	Excess of coalition S
N	Number of members of the EC
S	Coalition of users within the EC
$u_{i,k}^- / u_{i,k}^{dis-}$	Cost of member i in the cooperative/disagreement case for typical day k , €
$u_{j,k}^+ / u_{j,k}^{dis+}$	Profit of prosumer j in the cooperative/disagreement case for typical day k , €
$u_{i,k}^{incr-}$	Cost saving of member i for typical day k , €
$u_{j,k}^{incr+}$	Profit increase in prosumer j for typical day k , €
u_i^{incr}	Annual incremental economic benefit of member i , €
$v(S)$	Value of coalition S , €
w_k	Weight of typical day k
Subscripts and superscripts	
c	Consumer
dis	Disagreement case
exp	Export to the grid
i	Member of the EC
imp	Import from the grid
$incr$	Incremental
j	Prosumer
k	Typical day of the year
max	Maximum
$shift$	Shifted electricity demand
t	Hour of typical day
tr	Mid-market rate price
tr,b	P2P purchase
tr,s	P2P sale
$-/+$	Purchase/sale

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